

LUBNY, Ukraine

WMO: 333770

Lat: 50.0158N

Lon: 32.9867E

Elev: 158

StdP: 99.44

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-15.1	-21.5	0.6	-17.0	-18.7	0.7	-14.6	7.5	-4.0	6.4	-3.9	1.8	0	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	31.6	20.5	29.9	19.7	28.2	19.0	21.7	28.6	20.9	27.6	20.1	26.5	2.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.4	25.0	18.6	13.7	23.9	17.8	13.0	23.1	64.2	28.7	60.9	27.6	58.3	26.7	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.0	4.3	DB	-21.1	33.8	3.4	1.9	-23.5	35.2	-25.5	36.4	-27.4	37.5	-29.9	38.9	(4)
(5)			WB	-21.4	23.1	3.3	1.1	-23.8	23.9	-25.7	24.5	-27.6	25.1	-30.0	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-4.6	-3.3	1.7	9.9	15.9	19.4	21.3	20.4	14.8	8.2	2.4	-2.4	(6)	
	DBStd	10.30	5.47	5.62	4.46	4.29	4.47	3.64	3.09	3.63	4.09	4.42	4.67	5.30	(7)	
	HDD10.0	1852	452	373	257	53	7	0	0	0	7	88	231	386	(8)	
	HDD18.3	3802	710	606	514	254	103	32	8	19	119	315	478	644	(9)	
	CDD10.0	1377	0	0	1	50	188	282	351	321	151	32	2	0	(10)	
	CDD18.3	287	0	0	0	1	26	63	101	82	14	1	0	0	(11)	
	CDH23.3	2461	0	0	0	11	236	514	822	756	121	3	0	0	(12)	
	CDH26.7	735	0	0	0	2	52	131	257	269	25	0	0	0	(13)	
(14) Wind	WSAvg	2.0	2.2	2.3	2.3	2.1	1.8	1.8	1.7	1.6	1.7	1.8	2.2	2.3	(14)	
(15) Precipitation	PrecAvg	625	50	44	42	44	42	69	72	66	46	39	56	58	(15)	
	PrecMax	974	141	94	101	102	115	168	162	153	175	134	116	104	(16)	
	PrecMin	429	6	2	3	9	7	12	8	11	4	1	7	4	(17)	
	PrecStd	129	30	24	25	24	24	40	37	33	40	34	25	26	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	8.5	16.0	24.6	30.5	31.7	33.3	35.2	29.4	22.7	15.3	9.5	(19)	
		MCWB	5.1	5.8	8.7	14.2	18.4	19.8	21.2	20.0	17.9	15.0	12.4	8.7	(20)	
	2%	DB	4.5	5.9	12.7	21.5	28.0	29.8	31.4	32.0	26.8	19.8	12.0	7.0	(21)	
		MCWB	3.5	4.5	7.4	12.5	17.1	19.5	21.0	20.7	16.8	14.1	10.3	6.1	(22)	
	5%	DB	2.9	4.3	10.2	19.2	26.2	28.2	29.6	29.6	24.1	16.8	10.1	5.3	(23)	
		MCWB	2.1	3.2	6.2	11.7	16.8	19.1	20.2	19.8	16.0	13.0	8.9	4.7	(24)	
	10%	DB	1.8	2.9	7.9	17.2	23.8	26.5	27.8	27.4	21.8	14.5	8.5	3.7	(25)	
		MCWB	1.2	1.9	5.0	10.6	15.9	18.5	19.8	18.7	15.1	11.6	7.5	3.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	6.5	10.0	15.3	20.0	21.9	23.0	22.9	19.5	17.1	12.4	8.8	(27)	
		MCDB	6.2	7.8	14.5	23.2	27.2	28.1	29.9	30.1	26.8	20.0	14.7	9.3	(28)	
	2%	WB	3.6	4.8	8.1	13.6	18.6	20.8	22.0	21.4	18.0	14.9	10.6	6.3	(29)	
		MCDB	4.2	5.7	11.5	19.4	25.5	27.3	29.0	28.9	23.8	18.1	11.8	6.8	(30)	
	5%	WB	2.2	3.3	6.6	12.4	17.6	20.0	21.2	20.4	16.8	13.4	9.1	4.7	(31)	
		MCDB	2.7	4.1	9.4	17.4	24.3	26.4	27.7	27.8	22.3	16.2	9.8	5.3	(32)	
	10%	WB	1.3	2.1	5.2	11.5	16.5	19.1	20.4	19.5	15.7	11.9	7.6	3.2	(33)	
		MCDB	1.8	2.7	7.7	16.1	22.5	25.1	26.3	26.2	20.5	14.1	8.4	3.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	5.2	6.4	8.9	10.2	9.5	9.6	10.5	9.2	7.5	4.6	4.1	(35)	
		MCDBR	4.1	5.6	10.2	11.8	12.6	11.7	11.7	13.0	12.8	11.4	6.5	4.8	(36)	
	5% WB	MCWBR	3.6	4.5	6.1	5.6	5.0	4.6	4.4	4.7	5.5	6.4	5.1	4.4	(37)	
		MCDWR	3.7	5.2	9.1	10.0	11.4	10.3	10.3	11.4	11.1	9.6	5.8	4.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.314	0.344	0.376	0.441	0.408	0.400	0.429	0.428	0.399	0.363	0.346	0.317	(40)	
		taud	2.185	2.126	2.172	2.099	2.276	2.314	2.241	2.230	2.278	2.354	2.299	2.258	(41)	
	Edn at Noon	Edn at Noon	696	768	817	799	851	860	825	805	787	751	660	639	(42)	
		Edh at Noon	77	105	121	145	127	123	131	126	108	84	70	63	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.94	1.76	2.97	4.41	5.79	6.17	6.01	5.31	3.55	2.04	0.83	0.61	(44)	
	RadStd	RadStd	0.18	0.27	0.48	0.39	0.50	0.54	0.34	0.36	0.49	0.30	0.14	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.76	N/A	N/A	+1.06	N/A	N/A	N/A	-210	+145	+91	(46)
(47)	Regional (11 neighbors)	+0.73	N/A	N/A	+0.99	N/A	N/A	-140	-196	+116	+63	(47)

Nomenclature: See separate page